

driftsel

This is a manual for the use of driftsel software. driftsel stands for drift and selection. The theoretical framework of driftsel has been published by Ovaskainen et al. (2011), while a software paper has been written by Karhunen et al. (2013a). Additionally, the estimation of coancestry matrices used by Ovaskainen et al. (2011) has been explained in more detail by Karhunen and Ovaskainen (2012), and the so-called H test has been described by Karhunen et al. (2013b). In practice, driftsel is typically used in conjunction with RAFM which is used to estimate the coancestry matrices.

This manual is intended to help you to use driftsel. It is written in a style different from the official R documentation, as the standard documentation is not a very informative format in this case. From now on, we assume that you have successfully unzipped the files in the driftsel.zip and you have set your R in the same working directory.

What is driftsel good for? driftsel is good for evolutionary quantitative genetics, if you know enough R. It is good for your analyses, if you are prepared to do the brainwork required and understand the concepts (e.g. G matrices and posterior uncertainty). If you just want a simple summary statistic, it is probably easier to calculate a usual F_{ST} - Q_{ST} test. But in that way, you miss something essential: the joy and passion of Bayesian modelling. driftsel is genuinely multivariate and accounts for the correlations between different traits.

driftsel depends on `corpcor`, `MCMCpack` and `SparseM`.

First, the data

Type

```
load("demonstration.RData")
```

```
ls()
```

You should get

```
[1] "covars"      "genotypes"   "ped"         "traits"
```

This is an artificially created data set representing a full-sib study design. Type `head` and `is.matrix` for all of these matrices. *This is how you should format your data for use with driftsel*. Note that in R, data frame and matrix are two different things. In this example data set, `covars` is a set of environmental covariates, `ped` is the matrix representation of the pedigree, and `traits` are the traits to be analyzed. `genotypes` are the neutral genotypes used for calculating the coancestry matrix (see below). An investigator would typically have `covars`, `ped` and `traits` in a single Excel file. Thus, the use of driftsel requires some handiwork before you can carry on with the analysis.

The driftsel zip also involves three `.rda` files. These involve the data sets documented in `driftsel_manual_old.pdf` and are included for backward compatibility. This manual proceeds with the data from `demonstration.RData`.

Estimating the coancestry matrix

The coancestry matrix represents the neutral background to which the quantitative traits are contrasted. The estimation of coancestry matrix is, strictly speaking, RAFM analysis, but it is repeated here for completeness. It is done as

```
source("RAFM.r")  
  
afm <- do.all(genotypes, 50, 20, 2)  
  
theta <- afm$theta
```

Type

```
dim(theta)
```

You should get

```
[1] 3 3 15
```

The first dimensions (3 and 3) equal the number of subpopulations. The last dimension (15) is the MCMC sample size for `theta`. In practice, 15 is much too small, but we will use it for illustrative purposes. Alternatively, if you don't want to use RAFM, you can replace `theta` by some other posterior with same dimensions. However, in this case, you should be aware of the theoretical restrictions for `theta`, e.g. positive-definiteness (see Ovaskainen et al. 2011).

Finally, you typically want to estimate `theta` from individuals caught in the wild. If you use the offspring generation of your common-garden study to estimate `theta`, it can cause a bias in the within-population coancestries.

driftsel analysis

To run driftsel, type

```
source("driftsel.R")  
  
samp <- MH(afm$theta, ped, covars, traits, 100, 40, 2, alt=T)
```

Depending on your version of R and the packages, you can get a short or lengthy output. Then type

```
names(samp)
```

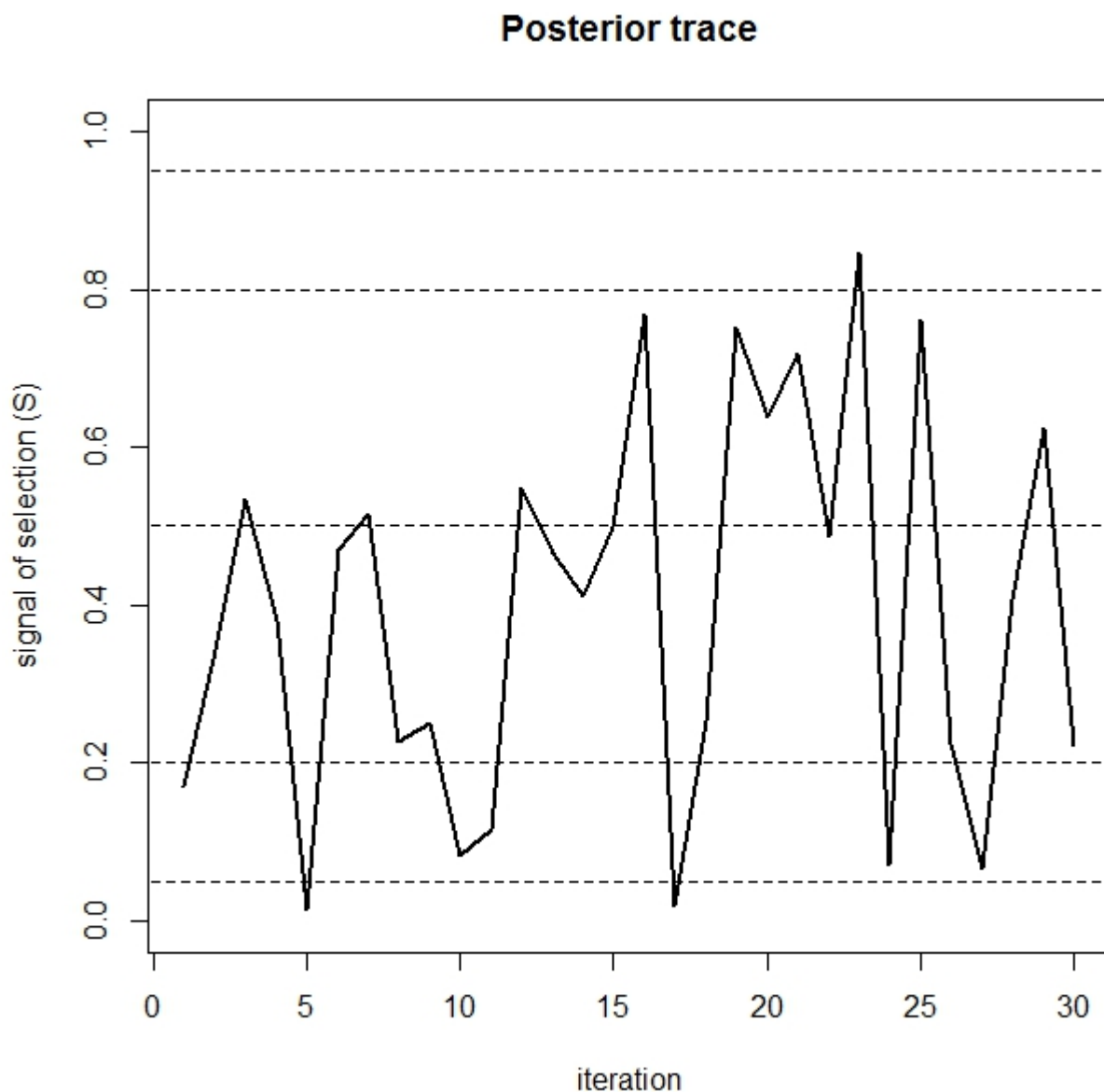
You should get

```
[1] "fixed.ef" "pop.ef"  "G"        "E"        "theta"
```

Each of these is the posterior sample of some parameter explained in Ovaskainen et al. (2011). The object `samp` involves a lot of information which could be used to write some interesting prose. However, the user typically wants to perform neutrality tests. To this end, type

```
neut.test(samp$pop.ef, samp$G, samp$theta, silent=F)
```

You should get some scalar number. This is the S test statistic of Ovaskainen et al. (2011). You also get something like



This is the posterior trace of S , of which the S test statistic is the mean. The trace of S can be evaluated as any other posterior sample. Here we see that the MCMC sample size is much too small (30). Type

`head(MH)`

You should get

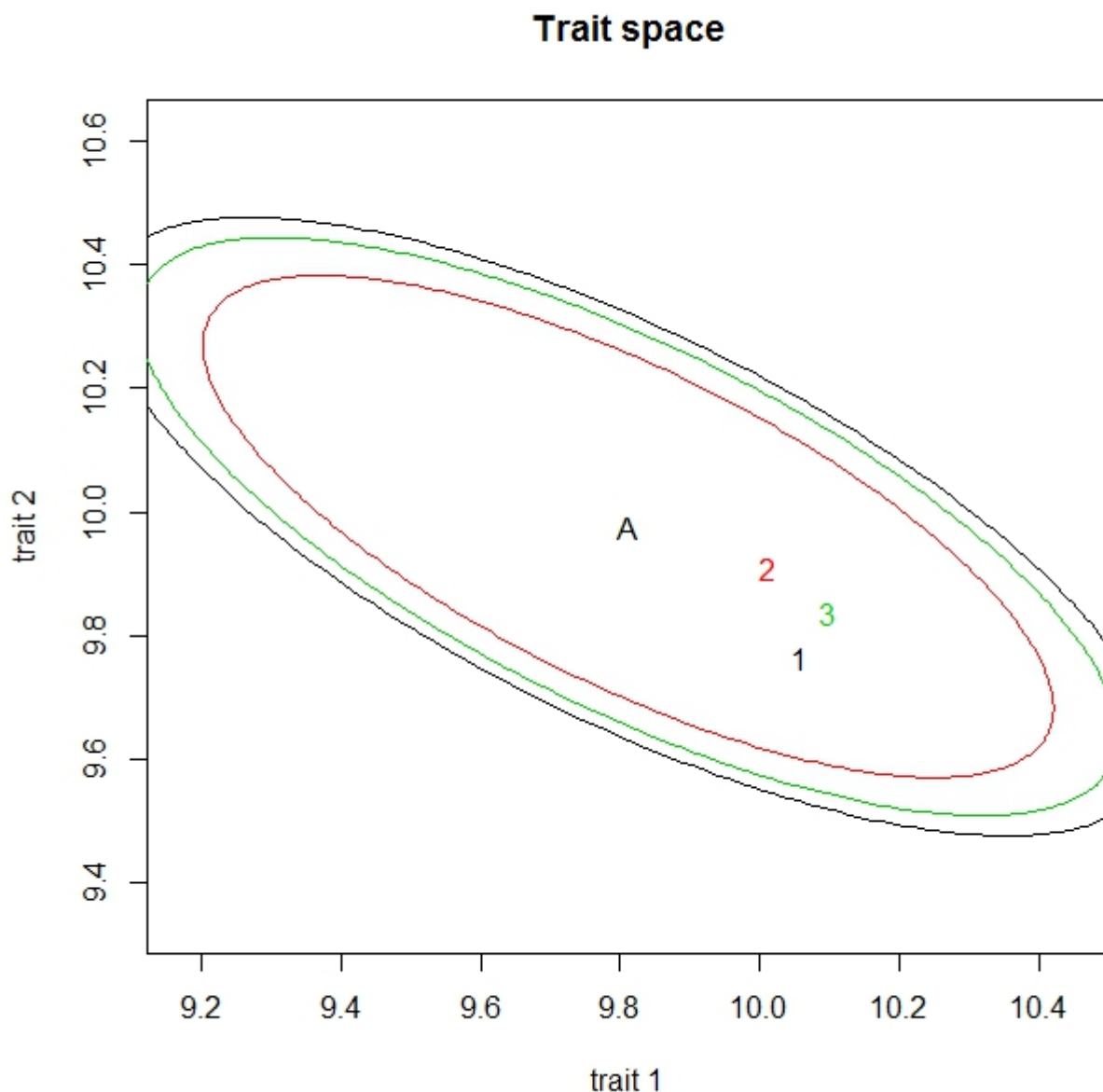
```
1 function (poster, ped, covars, traits, nmc, burnin, thin, blocks = NA,  
2     priors = NA, tmp = NA, binary = NA, alt = F)  
3 {  
4     library(corpcor)  
5     library(MCMCpack)  
6     library(SparseM)
```

The parameters `nmc`, `burnin` and `thin` are the ones that control the MCMC sampling. For example, `MH(afm$theta, ped, covars, traits, 10000, 5000, 5, alt=T)` would mean: analyze these data with total sample size 10,000, leave out 5,000 burn-in iterations, and take every 5th of the remaining iterations.

Finally, type

```
viz.traits(samp$fixed.ef, samp$pop.ef, samp$G, samp$theta, 1:2,  
main="Trait space")
```

You should get something like



This figure presents the population means in a trait space where A is the hypothetical ancestral population. The ellipses are the median drift distances. (This figure is different for each population, as each population has a different pattern of coancestry.) The farther the population means are from the central, the more likely divergent selection is. On the other hand, if the population means are very close to A, this may indicate stabilizing selection. The S test accounts for both phenomena.

In the above function call (`viz.traits(samp$fixed.ef, ...)`) the argument `1:2` declares the traits to be illustrated in `viz.traits`. For example, try replacing `1:2` by `c(1, 3)`.

The H test

The H test asks, if the traits are suspiciously correlated with the environment (Karhunen et al. 2013b). H stands for habitat. To perform the H test, first generate a dummy habitat:

```
pops <- 1:3      # population codes

rainfall <- c(0, 0, 111)    # random stuff

wind <- c(10, -4, -3)      # random stuff

habitat <- cbind(pops, rainfall, wind)
```

Then, run the test as

```
H.test(samp$pop.ef, samp$G, samp$theta, habitat, silent=F)
```

You get the value of the H test statistic and the trace of two Markov chains. Both should be well mixing. Finally, `habitat` concerns the natural habitat of the study populations, whereas `covars` (used earlier) concerns the conditions in the common-garden experiment. `habitat` has one row for each population, whereas `covars` has one row for each individual.

Custom analyses

Arguably, this manual presents the bare minimum you need to know to do driftsel analysis. Unfortunately, driftsel is no longer actively maintained, and we are slow to answer individual queries. However, we offer you the source code (`driftsel.r`) and the standard documentation (`driftsel_manual_old.pdf`). By using these two files, a skilled statistician should be able to further develop driftsel. To tap the whole potential of driftsel, you should read the standard documentation. For example, it is possible to include random effects in MH.

Known anomalies

The majority of problems with driftsel are caused by badly formatted data. There are two formats for feeding the predigree, `alt=F` and `alt=T`. These correspond to data sets `specimen` (or `demonstration`) and `specimen2`, respectively. These are explained in more detail in the old manual.

References

Karhunen M and Ovaskainen O (2012): Estimating population-level coancestry coefficients by an admixture F model. *Genetics* 192(2): 609-617.

Karhunen M, Merilä J, Leinonen T, Cano JM, Ovaskainen O (2013a): driftsel: an R package for detecting signals of natural selection in quantitative traits. *Molecular Ecology Resources* 13(4): 746-754.

Karhunen M, Ovaskainen O, Herczeg G and Merilä J (2013b): Bringing habitat information into statistical tests of local adaptation in quantitative traits: a case study of nine-spined sticklebacks. *Evolution* 68(2): 559-568.

Ovaskainen O, Karhunen M, Zheng C, Arias JMC, Merilä J and Spencer HG (2011): A new method to uncover signatures of divergent and stabilizing selection in quantitative traits. *Genetics* 189(2): 621-632.